

Storm-Gauge Forward Model

Spatial Decay and Intensity-to-Level Conversion

MKM Research Labs

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Contents

1 Executive Summary **4**

2 Model Purpose and Scope **4**
2.1 Purpose . . . 4
2.2 Scope . . . 5
2.3 Model Classification . . . 5

3 Mathematical Framework **5**
3.1 Haversine Distance . . . 5
3.2 Storm Track Interpolation . . . 5
3.3 Spatial Decay Kernels . . . 6
3.3.1 Gaussian Kernel . . . 6
3.3.2 Exponential Kernel . . . 6
3.3.3 Linear Kernel . . . 6
3.4 Intensity at Gauge . . . 6
3.5 Intensity-to-Level Conversion . . . 7
3.6 Temporal Response Dynamics . . . 7
3.7 Summary Statistics . . . 8

4 Input Parameters and Calibration **8**
4.1 Storm Parameters . . . 8
4.2 Gauge Configuration Parameters . . . 9
4.3 Model Hyperparameters . . . 9
4.4 Calibration Approach . . . 9
4.5 Data Formats . . . 9
4.5.1 Storm Track Data . . . 9
4.5.2 Gauge Configuration . . . 10

5 Implementation Details **10**
5.1 Software Architecture . . . 10
5.2 Computational Flow . . . 10
5.3 Numerical Considerations . . . 10
5.4 Dependencies . . . 10

6 Sensitivity Analysis **11**
6.1 Decay Kernel Selection . . . 11
6.2 Power-Law Exponent . . . 11
6.3 Storm Response . . . 12
6.4 Time Constants . . . 12
6.5 Footprint Radius . . . 13

7 Validation and Backtesting **13**
7.1 Unit Testing . . . 13
7.2 Benchmark . . . 13
7.3 Backtesting . . . 13

8 Model Limitations and Known Weaknesses **13**

9 Governance Validation Questions **14**
9.1 Q1 — Ronseal Test . . . 14
9.2 Q2 — Accuracy and Stability . . . 14
9.3 Q3 — Resource Usage . . . 14
9.4 Q4 — Override Controls . . . 14
9.5 Q5 — Output Consumption . . . 14
9.6 Q6 — Model Chaining . . . 14
9.7 Q7 — Weaknesses . . . 15
9.8 Q8 — Monitoring . . . 15
9.9 Q9 — Overall Risk . . . 15

10 Change History **16**

11 Document History **16**

1 Executive Summary

The Storm-Gauge Forward Model maps a storm's track, intensity profile and footprint to a water level time series at each river gauge. It has three stages:

1. **Spatial proximity:** the storm's position, interpolated linearly along its track at each time step, and its Haversine distance from the gauge.
2. **Intensity attenuation:** a Gaussian, exponential or linear decay kernel applied to the storm's intensity.
3. **Intensity-to-level conversion:** a power law gives each intensity a target level, and the gauge approaches it after a lag, rising and falling with separate time constants.

This is version 2.0, a revised model. The Wolfram benchmark of v1.3 (MKM-SG-001_wolfram_benchmark v0.1, 04-Nov-2026) found that the code computed its documented equations exactly. What the document said about them did not hold. v2.0 changes the model in three ways:

- **The rise has its own time constant,** $\tau_{\text{rise}} = 1\,\text{h}$. In v1.3 the rise rate was tied to the fall, at $3\Delta t/\tau_{\text{decay}}$, a 4-hour rise. The gauge reached only three-quarters of its target before the storm passed. A storm therefore needed a peak of 56 to reach the flood alert, where the calibration said 40, and never reached severe. Doubling τ_{decay} lowered the peak. Now a storm crossing the gauge reaches alert at peak 40.9, warning at 68.2 and severe at 90.7 (§3.5). τ_{decay} governs only the fall.
- **The run includes the recession.** v1.3 stopped at the storm's end, with the gauge still high, so durations and accumulation were cut short. v2.0 runs five fall constants further, with no forcing.
- **An unread input is removed.** `intensity_to_level_scale` was stored and never read. The constants move to `config.models` (R1).

Where the model is used. v1.3 described the model as feeding `gaugets/`, the hazard curves and the flood risk model. It does not, and did not. Nothing in the platform imports it. `gaugets/` is written by a scripted flood wave (`port/rand/shared/gauge/gaugets_random`). The model is a standalone forward model, used for analysis and as the reference for the gauge response that MKM-SS-001 re-implements (§2.2).

The model is implemented in Python with only the standard library's `math` module, so it is portable and easy to audit.

2 Model Purpose and Scope

2.1 Purpose

The model computes the water level response at each gauge in a catchment to a parameterised storm. It supports:

- analysis of how a gauge responds to storm characteristics: intensity, track proximity, footprint width and kernel shape;
- the spatial pattern of flooding across a catchment's gauges for one storm;
- a reference for the lagged, asymmetric gauge response that other models re-implement.

2.2 Scope

- **In scope:** track interpolation; Haversine distance; the decay kernels; the intensity-to-level transfer; the lagged rise and fall; and summary statistics (peak level, durations above each stage, accumulation).
- **Out of scope:** storm generation (`storm_factory.py` builds simple storms for the model's own use); hydrodynamic solvers; property depth–damage; PRS pricing.
- **Upstream:** `Storm` objects, from `create_storm` or built directly; `GaugeConfig` objects, from the gauge portfolio CDM through `GaugeConfig.from_gauge_portfolio`.
- **Downstream: none in production.** No module in `src`, `tools` or `scripts` imports the package. The gauge time series in `gaugets/` come from `port/src/gauge/gaugets.py`, which draws a scripted wave from `gaugets_random`. MKM-SS-001's sequence response (`storm_multi/models/sequence_response.py`) re-implements a lagged, asymmetric rise and fall of the same kind; it does not call this code.

2.3 Model Classification

Under the firm's model risk management framework this is a *Tier 2 quantitative model*. It is deterministic given its inputs. Because no production model consumes its output, an error in it cannot at present reach a price, a risk figure or a report. That would change if it were wired into `gaugets/`, and this document would have to change with it.

3 Mathematical Framework

3.1 Haversine Distance

The distance between a gauge at (ϕ_g, λ_g) and a storm position (ϕ_s, λ_s) is computed on a sphere of radius $R = 6{,}371\,\text{km}$:

$$a = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos\phi_g\,\cos\phi_s\,\sin^2\left(\frac{\Delta\lambda}{2}\right) \tag{1}$$

$$d = 2R\,\arctan\left(\frac{\sqrt{a}}{\sqrt{1-a}}\right) \tag{2}$$

where $\Delta\phi = \phi_s - \phi_g$ and $\Delta\lambda = \lambda_s - \lambda_g$, in radians. The code evaluates Eq. 2 with `math.atan2`, which equals $\arcsin\sqrt{a}$ without its loss of accuracy as $a \to 1$.

3.2 Storm Track Interpolation

A storm is an ordered sequence of track points $\{(\lambda_i, \phi_i, t_i, I_i)\}_{i=1}^{N}$: longitude, latitude, hours since the storm started, and intensity on a $[0, 100]$ scale. At time t the model finds the bracketing points with $t_{\text{prev}} \le t < t_{\text{next}}$ and the fraction

$$\alpha = \text{clamp}\left(\frac{t - t_{\text{prev}}}{t_{\text{next}} - t_{\text{prev}}},\ 0,\ 1\right) \tag{3}$$

and interpolates linearly:

$$\lambda_s(t) = \lambda_{\text{prev}} + \alpha\,(\lambda_{\text{next}} - \lambda_{\text{prev}}) \tag{4}$$

$$\phi_s(t) = \phi_{\text{prev}} + \alpha\,(\phi_{\text{next}} - \phi_{\text{prev}}) \tag{5}$$

$$I_s(t) = I_{\text{prev}} + \alpha\,(I_{\text{next}} - I_{\text{prev}}) \tag{6}$$

Before the first track point the storm is at that point; after the last, at the last. After the storm's duration T its intensity is zero (§3.6).

3.3 Spatial Decay Kernels

The storm's footprint radius F (km) normalises the distance,

$$\tilde{d} = \frac{d}{F}, \tag{7}$$

and one of three kernels attenuates the intensity:

3.3.1 Gaussian Kernel

$$k_{\text{Gauss}}(\tilde{d};\,\sigma) = \exp\!\left(-\frac{\tilde{d}^{\,2}}{2\sigma^2}\right) \tag{8}$$

with bandwidth σ (default 0.5). It is the default kernel. At $\tilde{d} = \sigma$ the intensity is 60.7% of its on-track value, and at 2σ it is 13.5%.

3.3.2 Exponential Kernel

$$k_{\text{Exp}}(\tilde{d};\,\lambda) = \exp\!\left(-\frac{\tilde{d}}{\lambda}\right) \tag{9}$$

with decay length λ (default 0.5). It is sharper at the centre than the Gaussian, and has heavier tails.

3.3.3 Linear Kernel

$$k_{\text{Lin}}(\tilde{d};\,r) = \max\!\left(0,\;1 - \frac{\tilde{d}}{r}\right) \tag{10}$$

with cut-off r (default 0.5). Intensity is zero at and beyond $\tilde{d} = r$.

Boundary condition. For every kernel $k(0) = 1$: a gauge on the track receives the full intensity. The Gaussian and exponential kernels tend to zero with distance; the linear kernel reaches it.

3.4 Intensity at Gauge

The intensity a gauge experiences at time t is

$$I_{\text{gauge}}(t) = I_s(t) \times k\Big(\tilde{d}(t)\Big)\,, \tag{11}$$

with $I_s(t)$ from Eq. 6 and $\tilde{d}(t)$ the normalised distance to the storm's position at t. Intensity and footprint are assumed separable (§8, item 3).

3.5 Intensity-to-Level Conversion

Each intensity has a target level. For a gauge with base level L_{base}, historical high L_{high} (or $1.1 \times L_{\text{severe}}$ if it has none) and sensitivity s:

$$L_{\text{range}} = L_{\text{high}} - L_{\text{base}} \tag{12}$$

$$\Delta L(I) = \left(\frac{I}{100}\right)^{\gamma} \times L_{\text{range}} \times s, \qquad \Delta L(I) = 0 \text{ for } I \le 0, \tag{13}$$

with $\gamma = 0.8$. The sub-linear exponent gives moderate intensities a proportionally larger target than a linear map would.

The calibration bands. At Table 2's default stages the target reaches each stage at a fixed intensity, whatever the alert level: flood alert at $I = 38.7$, flood warning at 64.7, severe at 86.0 and the historical high at 100. A target is only reached if the intensity is held for long enough. A storm passes, so it must peak higher. A default storm (Table 1) crossing directly over the gauge reaches:

- flood alert at peak intensity 40.9;
- flood warning at 68.2;
- severe at 90.7.

These storm figures are the ones v2.0's rise constant was chosen to give. The design intent is alert near 40, warning in 60–70 and severe at 85 or above (Table 8). A gauge with its own historical high has its own bands.

3.6 Temporal Response Dynamics

The water level at step i is

$$L(t_i) = L_{\text{base}} + C(t_i), \tag{14}$$

where C is the contribution. With time step Δt (0.5 h) and lag τ_{lag} (2 h), the target at step i is ΔL of the lagged intensity

$$I_{\text{lag}}(t_i) = I_{\text{gauge}}\left(t_{\max(0,\, i - \lfloor \tau_{\text{lag}}/\Delta t \rfloor)}\right), \tag{15}$$

and the contribution approaches it with a rise constant τ_{rise} (1 h) and a fall constant τ_{decay} (12 h):

$$C(t_i) = C(t_{i-1}) + \left[\Delta L_{\text{target}} - C(t_{i-1})\right] \times \begin{cases} \min(1,\ \Delta t/\tau_{\text{rise}}) & \text{if } \Delta L_{\text{target}} > C(t_{i-1}) \\ \min(1,\ \Delta t/\tau_{\text{decay}}) & \text{otherwise,} \end{cases} \tag{16}$$

from $C = 0$. The gauge rises twelve times faster than it falls.

The run. The grid is $t_i = i\,\Delta t$ for $i = 0, \ldots, \lfloor (T + 5\tau_{\text{decay}})/\Delta t \rfloor$. After the storm's duration T there is no forcing. The five fall constants that follow let the contribution fall to e^{-5}, under 1%, of where it started falling.

Lag between the peaks. Peak level follows peak intensity by τ_{lag} plus the time the rising level takes to meet the falling target. For a default storm that is 2.5 h (Table 8).

3.7 Summary Statistics

Over the whole run, recession included:

- **Peak level** $L_{\text{peak}} = \max_i L(t_i)$, and its time.
- **Peak exceedance** $E_{\text{peak}} = L_{\text{peak}} - L_{\text{alert}}$, negative if the gauge does not reach alert.
- **Duration above each stage** θ (alert, warning, severe):

$$D_\theta = \sum_{i:\, L(t_i) \geq \theta} \Delta t \tag{17}$$

- **Accumulation** above the alert level, in metre-hours:

$$A = \sum_i \max\!\big(0,\ L(t_i) - L_{\text{alert}}\big)\, \Delta t \tag{18}$$

- **Peak intensity** $I_{\text{peak}} = \max_i I_{\text{gauge}}(t_i)$, and its time.

4 Input Parameters and Calibration

4.1 Storm Parameters

Table 1: Storm input parameters (defaults of `create_storm`)

Parameter	Symbol	Units	Default
Track points	$\{(\lambda_i, \phi_i, t_i, I_i)\}$	deg, h, —	25 points
Peak intensity	I_{peak}	0–100	50.0
Footprint radius	F	km	40.0
Duration	T	hours	24.0
Decay kernel	—	enum	Gaussian
Decay parameter	σ, λ or r	—	0.5
Intensity profile	—	enum	Triangular

The peak intensity and profile shape the track's intensities when `create_storm` builds it; the model reads only the track. A `Storm` also carries a speed, which the model does not read.

4.2 Gauge Configuration Parameters

Table 2: Gauge configuration parameters (defaults of `from_gauge_portfolio`)

Parameter	Symbol	Units	Default
Base level	L_{base}	m	$0.35 \times L_{\text{alert}}$
Flood alert	L_{alert}	m	4.0
Flood warning	L_{warn}	m	$1.33 \times L_{\text{alert}}$
Severe warning	L_{severe}	m	$1.58 \times L_{\text{alert}}$
Historical high	L_{high}	m	optional
Sensitivity	s	—	1.0

4.3 Model Hyperparameters

Table 3: Model hyperparameters (`config.models`, `_stormgauge.py`)

Parameter	Symbol	Units	Default
Time resolution	Δt	hours	0.5
Response lag	τ_{lag}	hours	2.0
Rise time constant	τ_{rise}	hours	1.0
Fall time constant	τ_{decay}	hours	12.0
Recession run-on	—	τ_{decay}	5
Power-law exponent	γ	—	0.8
Historical high, where none	—	$\times L_{\text{severe}}$	1.1
Earth radius	R	km	6,371

4.4 Calibration Approach

The parameters are design choices, not fitted values.

- $\gamma = 0.8$ and the stage multiples of Table 2 set the steady-state bands of §3.5.
- $\tau_{\text{rise}} = 1\,\text{h}$ was chosen in v2.0 so that a storm crossing the gauge reaches the bands those choices intend.
- $\tau_{\text{lag}} = 2\,\text{h}$ and $\tau_{\text{decay}} = 12\,\text{h}$ are unchanged from v1.3.

v1.3 described the parameters as calibrated against Environment Agency records for the Thames. No such calibration is held in the repository, and v2.0 does not repeat the claim. Calibrating them to observed hydrographs is the first item of §8.

The gauge sensitivity s (default 1.0) can be set per gauge, for catchment characteristics such as urbanisation and soil.

4.5 Data Formats

4.5.1 Storm Track Data

A storm's track is a list of `TrackPoint` objects:

```
@dataclass
class TrackPoint:
    longitude: float    # Degrees (WGS84)
```

```
    latitude: float      # Degrees (WGS84)
    time_hours: float    # Hours since storm start
    intensity: float     # 0-100 intensity scale
```

4.5.2 Gauge Configuration

`GaugeConfig.from_gauge_portfolio()` reads a gauge's stages and coordinates from its gauge portfolio CDM entry, with the defaults of Table 2 where a stage is missing.

5 Implementation Details

5.1 Software Architecture

The package `src/models/stormgauge/` holds:

- `data_structures.py`: the enumerations `DecayKernel` and `IntensityProfile`, and the dataclasses `TrackPoint`, `Storm`, `GaugeConfig` and `GaugeResponse`;
- `forward_model/`: `StormGaugeModel` (`_core.py`), with distance, decay, interpolation and the power law in `_intensity.py` and the response in `_response.py`;
- `storm_factory.py`: `create_storm` and `load_gauges_from_portfolio`.

Its parameters are in `config/models/_stormgauge.py`.

5.2 Computational Flow

`StormGaugeModel.compute_response(storm, gauge)`:

1. builds the grid of §3.6, from 0 to $T + 5\tau_{\text{decay}}$;
2. computes the intensity at the gauge at each step (Eqs. 3–11), zero after T;
3. applies the lagged target and Eq. 16;
4. extracts the statistics of §3.7;
5. returns a `GaugeResponse` with the series and the statistics.

`compute_all_responses(storm, gauges)` runs each gauge in turn. The cost is $\mathcal{O}(N_t \cdot N_{\text{track}})$ per gauge.

5.3 Numerical Considerations

- The Haversine uses `math.atan2` (§3.1).
- α is clamped to $[0, 1]$, so the storm never leaves its track.
- Both smoothing rates are clamped by $\min(1, \cdot)$, so a large step cannot overshoot its target.
- Arithmetic is 64-bit floating point. A kernel value below the smallest double, far outside the footprint, is zero.

5.4 Dependencies

The Python standard library (`math`, `dataclasses`, `enum`, `typing`) and `config.models`.

6 Sensitivity Analysis

Table 4: Spatial decay factor by kernel type (footprint 100 km, decay parameter 1.0).

Distance (km)	Gaussian	Exponential	Linear
0	1.00	1.00	1.00
25	0.9692	0.7788	0.7500
50	0.8825	0.6065	0.5000
75	0.7548	0.4724	0.2500
100	0.6065	0.3679	0.0000
150	0.3247	0.2231	0.0000
200	0.1353	0.1353	0.0000

Table 5: Target contribution ΔL (m, Eq. 13) by intensity and gauge sensitivity ($L_{\text{base}} = 1.5$, $L_{\text{high}} = 6.0$ m).

s	$I = 20$	$I = 40$	$I = 60$	$I = 80$	$I = 100$
0.8	0.9930	1.73	2.39	3.01	3.60
1.0	1.24	2.16	2.99	3.76	4.50
1.2	1.49	2.59	3.59	4.52	5.40

6.1 Decay Kernel Selection

The kernel matters for storms that do not pass over the gauge. One footprint radius away ($\tilde{d} = 1.0$):

Table 6: Decay factor at $\tilde{d} = 1.0$ for each kernel (default parameters)

Kernel	$k(1.0)$	Relative intensity
Gaussian ($\sigma = 0.5$)	$\exp(-2) \approx 0.135$	13.5%
Exponential ($\lambda = 0.5$)	$\exp(-2) \approx 0.135$	13.5%
Linear ($r = 0.5$)	$\max(0, 1-2) = 0.0$	0.0%

The Gaussian and exponential kernels coincide here for $\sigma = \lambda = 0.5$ and nowhere else. The linear kernel gives nothing beyond $\tilde{d} = r = 0.5$.

6.2 Power-Law Exponent

Table 7: Normalised level contribution $\Delta L / L_{\text{range}}$ at $I = 50$ for varying γ

γ	$\Delta L / (s \cdot L_{\textbf{range}})$
0.5	$(0.5)^{0.5} \approx 0.707$
0.8 (default)	$(0.5)^{0.8} \approx 0.574$
1.0	$(0.5)^{1.0} = 0.500$
1.5	$(0.5)^{1.5} \approx 0.354$

A 25% change in γ moves the target at $I = 50$ by +14.9% (down to 0.6) and −12.9% (up to 1.0). Across $I = 40$–60 the change is 10–20%. The effect grows as the intensity falls.

6.3 Storm Response

Table 8: Response of a default storm (Table 1) crossing the Thames at 51.5°N, at a gauge with Table 2's default stages ($L_{\text{alert}} = 4\,\text{m}$) on the track and 0.1 and 0.2° north of it. Lag is peak level less peak intensity, in hours; durations in hours; accumulation in metre-hours.

Peak	North (°)	I_{peak}	L_{peak} (m)	Lag	D_{alert}	D_{warn}	D_{severe}	A
40	0.0	40.0	3.96	2.5	0.0	0.0	0.0	0.00
40	0.1	34.3	3.66	2.5	0.0	0.0	0.0	0.00
40	0.2	21.6	2.96	2.5	0.0	0.0	0.0	0.00
60	0.0	60.0	4.94	2.5	10.5	0.0	0.0	6.20
60	0.1	51.4	4.52	2.5	8.0	0.0	0.0	2.68
60	0.2	32.3	3.56	2.5	0.0	0.0	0.0	0.00
80	0.0	80.0	5.85	2.5	14.5	6.5	0.0	16.17
80	0.1	68.5	5.33	2.5	12.5	1.0	0.0	10.24
80	0.2	43.1	4.11	2.5	3.5	0.0	0.0	0.29
100	0.0	100.0	6.72	2.5	17.5	10.5	5.0	27.36
100	0.1	85.7	6.10	2.5	15.5	8.0	0.0	19.27
100	0.2	53.9	4.65	2.5	8.5	0.0	0.0	3.63

The lag between the peaks is the same in every case, 2.5 h: the storms share one profile, and the gauge's distance scales the forcing without reshaping it. Off the track the storm is weaker: 0.2° (22 km) north, a peak-100 storm reaches alert but not warning.

6.4 Time Constants

Table 9: Sensitivity to the time constants, on the peak-100 storm crossing the gauge; the others at their defaults.

Constant	Hours	L_{peak} (m)	Lag (h)	D_{alert} (h)	A (m h)
τ_{lag}	1	6.72	1.5	17.5	27.36
τ_{lag}	2	6.72	2.5	17.5	27.36
τ_{lag}	4	6.72	4.5	17.5	27.36
τ_{rise}	0.5	6.95	2.0	18.5	30.66
τ_{rise}	1	6.72	2.5	17.5	27.36
τ_{rise}	2	6.31	3.0	16.0	21.53
τ_{rise}	4	5.53	4.0	13.0	12.17
τ_{decay}	6	6.72	2.5	13.5	21.38
τ_{decay}	12	6.72	2.5	17.5	27.36
τ_{decay}	24	6.72	2.5	26.0	38.07

- τ_{lag} moves the timing one for one and leaves the peak alone.
- τ_{decay} leaves the peak alone and lengthens the recession: doubling it from 12 to 24 h takes the time above alert from 17.5 to 26 h.
- τ_{rise} sets how far the gauge gets before the storm passes. At 4 h, v1.3's effective value, the peak-100 storm's peak is 1.2 m lower.

6.5 Footprint Radius

The footprint F scales the reach of the storm: doubling it doubles the distance at which a given decay is reached. It decides which gauges a storm affects, and so drives the spatial pattern of flooding across a catchment.

7 Validation and Backtesting

7.1 Unit Testing

The package's tests (`tests/models/stormgauge/`) verify:

- the Haversine against known distances, and its symmetry;
- the kernels' boundary conditions: $k(0) = 1$, and the linear kernel zero for $\tilde{d} \geq r$;
- that a higher intensity gives a higher target, and zero intensity none;
- no forcing after the storm, and a completed recession;
- that τ_{decay} does not slow the rise, and lengthens the time above alert;
- that a held intensity brings the level to its Eq. 13 target.

7.2 Benchmark

The Wolfram oracle benchmark (MKM-SG-001_wolfram_benchmark) checks the code against a 50-digit implementation of this document's equations and against Wolfram's own. It also tests this document's figures and statements. Its v0.1, against v1.3, found the shortcomings this version corrects.

7.3 Backtesting

No backtest is on record. v1.3 reported peak errors within ± 0.3 m and timing within ± 1 h against Environment Agency records. It also reported cross-checks against GEV fits. Nothing in the repository runs the model against gauge records, so those figures cannot be reproduced, and v2.0 does not repeat them. A backtest against observed Thames hydrographs is the condition of Q9.

8 Model Limitations and Known Weaknesses

1. **Uncalibrated.** γ, τ_{lag}, τ_{rise} and τ_{decay} are design choices. None has been fitted to observed hydrographs.
2. **No antecedent conditions.** Soil moisture, groundwater and preceding rain are ignored: a storm always produces the same response. MKM-SS-001 composites sequences of storms with partial recovery between them.
3. **Separable intensity and decay.** The footprint's shape does not vary with intensity.
4. **Static sensitivity.** s does not vary with level or season.
5. **Spherical Earth.** Against the WGS84 ellipsoid the sphere understates distances in the UK by up to 0.32%, east–west, and by 0.06% north–south.
6. **One rise and one fall constant** for every gauge, whatever its catchment.
7. **No tidal interaction** at estuarine or coastal gauges.
8. **One footprint per storm,** where real storms have several bands.

9. **Linear track interpolation** in latitude and longitude rather than along great circles.

9 Governance Validation Questions

The following nine questions are mandated by Chapter 5 of the *Model Risk Governance for Vendors* handbook and must be explicitly addressed for every model in the inventory. Response status is tracked in the Model Governance Dashboard (MRC portal).

9.1 Q1 — Ronseal Test

"Does the model do what it says on the tin?"

Yes, as v2.0 describes it. It maps a storm to a water level series and summary statistics at each gauge, and the Wolfram benchmark confirms it computes this document's equations. v1.3 also said it supplied the platform's gauge time series, which it does not (§2.2).

9.2 Q2 — Accuracy and Stability

"Does the model provide timely, accurate and stable results?"

It is deterministic and numerically exact to double precision. Its accuracy against real gauges is unknown: no backtest is on record (§7.3).

9.3 Q3 — Resource Usage

"Does the model consume excessive computational resources?"

No. A 48-hour storm with 25 track points takes 0.3 ms per gauge, recession included, and 25 gauges take 8 ms.

9.4 Q4 — Override Controls

"Are there controls over manual overrides?"

The per-gauge sensitivity s comes from gauge configuration data. The other parameters are in `config.models` under version control, and can be passed to `StormGaugeModel` for analysis.

9.5 Q5 — Output Consumption

"How is the output consumed for risk decision-making?"

It is not. No production model, report or screen consumes the model's output. It is used for analysis, and as the reference shape for MKM-SS-001's sequence response.

9.6 Q6 — Model Chaining

"Is the model's output used as input to another model?"

No. Its inputs are storms built by `create_storm` or directly, and gauges from the gauge portfolio CDM. MKM-SS-001 re-implements a response of the same kind; it does not call this code.

9.7 Q7 — Weaknesses

"Does the model have weaknesses and limitations of use?"

Yes, nine (§8). The most material is that its parameters are uncalibrated.

9.8 Q8 — Monitoring

"What type of monitoring is required?"

While nothing consumes its output, a re-run of the benchmark on any change to the package or its parameters. If it is wired into production, the backtest of Q9 and an annual comparison against Environment Agency records.

9.9 Q9 — Overall Risk

"Is the overall model risk acceptable?"

Acceptable for its present use, which reaches no decision. Before its output is used in production it needs calibration and a backtest against observed Thames hydrographs, and this document must be revised to say where the output goes.

10 Change History

v2.0 (04-Nov-2026). After the Wolfram benchmark of v1.3:

- a rise time constant of its own, $\tau_{\text{rise}} = 1\,\text{h}$, in place of $3\Delta t/\tau_{\text{decay}}$;
- the run continues $5\tau_{\text{decay}}$ past the storm's end, with no forcing, so durations and accumulation include the recession;
- `intensity_to_level_scale`, never read, removed; the constants moved to `config.models`;
- the scope, calibration, backtesting and governance answers corrected to what the model is and where it is used;
- the γ sensitivity and the sphere's error restated as computed;
- Table 5 now shows sensitivity, and Tables 8–9 the storm response and the time constants.

11 Document History

Date	Version	Description	Author
13-Feb-2026	1.0	Initial documentation	D. Kelly
18-Feb-2026	1.1	Added Section 9: Governance Validation Questions (Chapter 5)	D. Kelly
26-Feb-2026	1.2	Added Delta Engine model to inventory; updated models in scope	D. Kelly
11-Mar-2026	1.3	Antecedent conditions partly addressed by MKM-SS-001	D. Kelly
04-Nov-2026	2.0	Revised model after the Wolfram benchmark: own rise constant, recession, unread scale removed; scope and governance corrected	D. Kelly